

Aditya Kumar Joshi

+91 7976158533

Gmail

LinkedIn

Github

LeetCode

Portfolio Website

Work Experience

Machine Learning Engineer (Intern)

May 2025 – July 2025

Technouniverse Softtech India Pvt. Ltd., Indore, Madhya Pradesh

- Developed an MVP for a Smart Attendance Marking System using Face Detection and Recognition, integrating Deep Learning and Computer Vision (OpenCV, Transformers) to automate real-time identity verification.
- Assisted in designing and developing user interfaces and backend services using React and FastAPI for seamless system integration.

Full Stack Developer

MAY 2024–April 2025

IES IPS Academy, Indore, Madhya Pradesh

- Developed and maintained the College Management System (CMS) built on PHP CodeIgniter.
- Designed and implemented the Faculty Development Program (FDP) and Research Paper Management modules.

Skills

Programming Languages: HTML, CSS, JavaScript, Python

Frameworks & Libraries: React Js, FastAPI, PyTorch, Hugging Face Transformers, OpenCV, DeepFace, NumPy, Pandas

Databases: MySQL, MongoDB,

Tools & Platforms: VS Code, Git, Postman, Google Colab, Kaggle, Google Sheets

Machine Learning & AI: Supervised & Unsupervised Learning (Linear/Logistic Regression, XGBoost), Natural Language Processing (NLP), LLMs, Retrieval-Augmented Generation (RAG), Agentic AI (LangChain, LangGraph), vector embeddings, Model Evaluation & Model Fine-Tuning, Computer Vision (YOLO-based Object Detection, Image/Video Analysis).

Projects

1. SecureVision X - Agentic AI-Based Weapon Detection System *YOLOv8, FastAPI, LangGraph, Gemini, React*
Trained a custom YOLOv8 weapon detector achieving **98.8% mAP@0.5, 90.4% mAP@0.5:0.95, 98.8% precision, and 96.9% recall** on 275 test images.

Engineered a **multi-agent security pipeline** using LangGraph, Gemini 2.5 Flash Vision, and Chroma RAG for risk assessment, emergency SOP retrieval, and automated notifications, with a **FastAPI + React** interface supporting image/video detection and real-time SSE workflow updates.

2. Mathematical Equation generator

NLP, mT5, Transformer

Fine-tuned an mT5 model on **12,744 Hindi math word problems** to generate symbolic equations from natural-language questions, achieving **97.73% exact-match accuracy** on 1,412 validation samples. Achieved **99.56% BLEU-1** with 1,380/1,412 correct predictions using 15-epoch training and validation-loss-based model selection.

3. HazardHawk AI

YOLOv11, XGBoost, FastAPI, React

Developed an end-to-end surveillance system for **Fall, Sit, Stand, and Person detection** in images, videos, and IP-camera streams using YOLOv11, achieving **88.3% mAP@50, 66.7% mAP@50-95, 85.1% precision, and 80.5% recall**. Implemented an **XGBoost secondary verification model** using bounding-box spatial features, achieving **98.2% accuracy and 0.98 F1-score** for fall detection, with FastAPI + React for real-time visualization and analytics.

4. Smart Attendance Marking System

FaceNet, Deepface, FastAPI, React

Built a real-time face recognition attendance system with snapshot logging and real-time stats. Integrated backend (FastAPI) and frontend (React) for seamless classroom/meeting usage. Future integration planned with motions so no fake attendances.

Education

Bachelor of Technology in CSE

Graduation: May 2026

IES IPS Academy, Indore, Madhya Pradesh

CGPA: 7.8

Certifications & Extracurricular Activities

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- SMART INDIA HACKATHON 2025 WINNER
- Winner (Phase 1) & Runner-up (Phase 2), UDAAN Project Competition, IPS Academy.